

lab6

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```
library(haven)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##   filter, lag
```

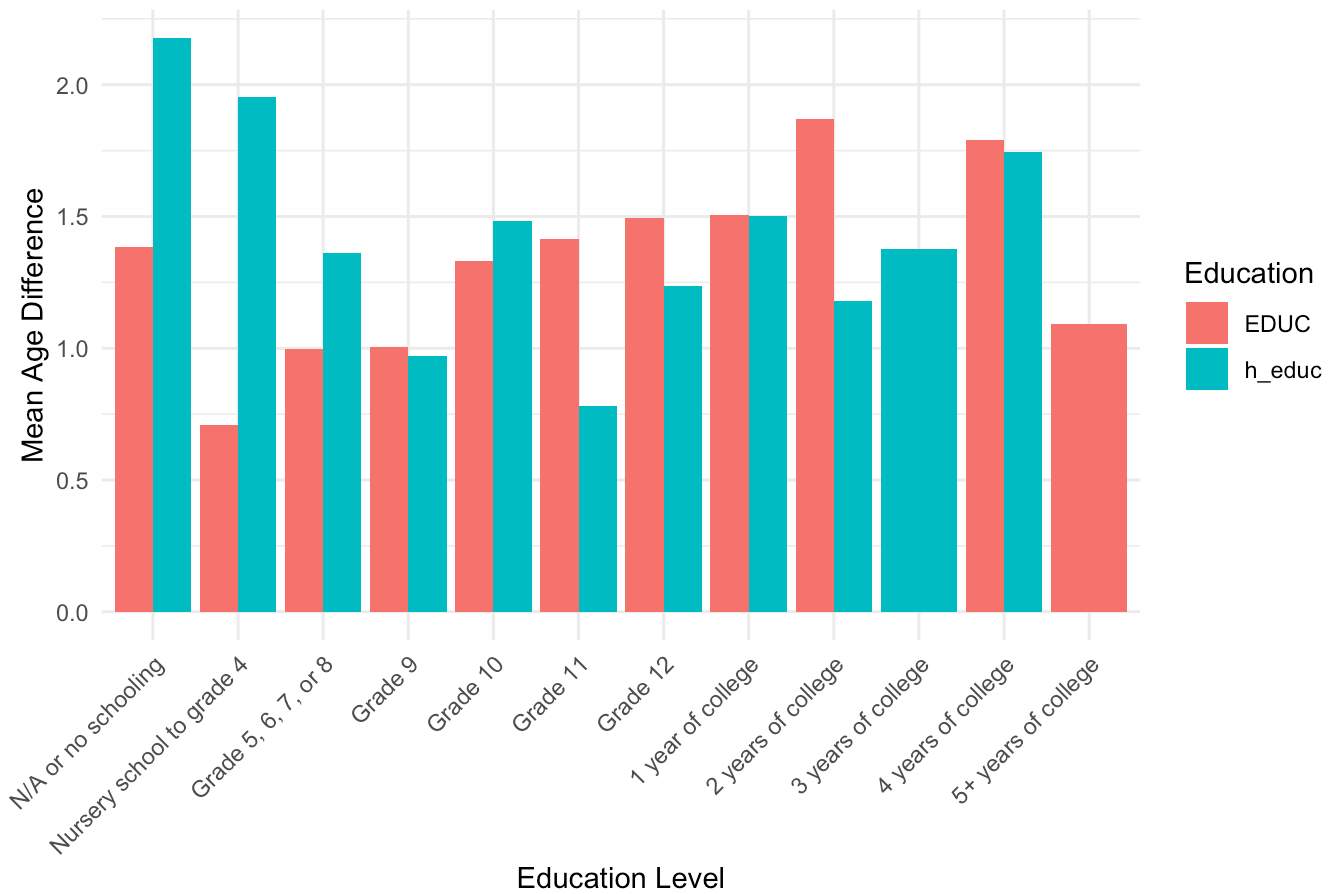
```
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(forcats)
library(tibble)
library(ggplot2)
library(class)
library(tidyr)
setwd("~/Documents/Masters/Econometrics/data")# your directory structure will be different
load("ACS_2021_couples.RData")
```

This lab investigates whether educational attainment and immigrant status influence the average spousal age difference in U.S. married couples using data from the 2021 American Community Survey (ACS).

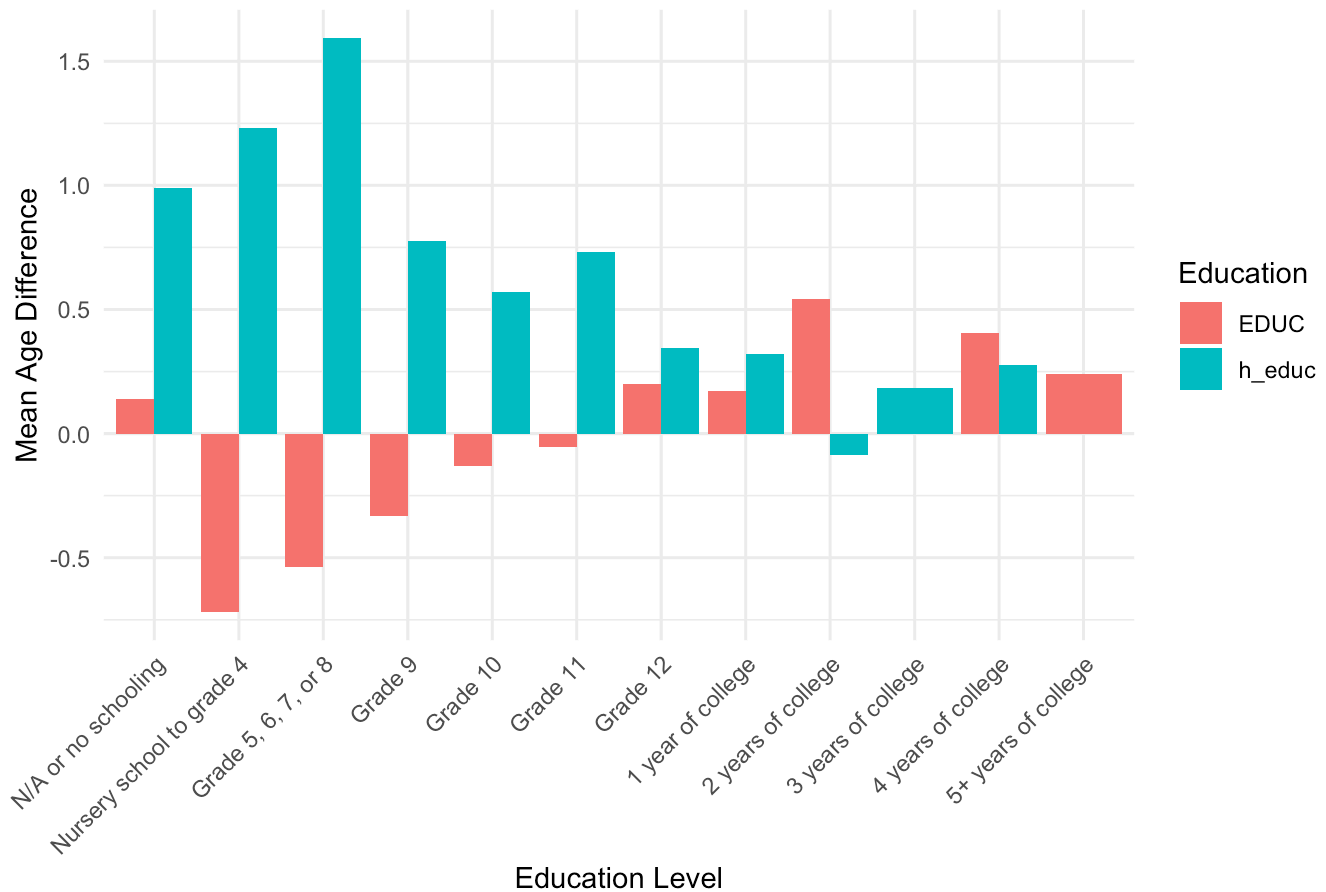
```
## `summarise()` has grouped output by 'Education_Type'. You can override using
## the `groups` argument.
```

Mean Age Difference by Education(Head of Household vs Spouse) Not Born in l



```
## `summarise()` has grouped output by 'Education_Type'. You can override using  
## the `.groups` argument.
```

Mean Age Difference by Education(Head of Household vs Spouse) Born in US



```
anova_EDUC = aov(Age_DIFF ~ EDUC, data = acs2021_couples_spouse)
summary(anova_EDUC)
```

```
##              Df    Sum Sq Mean Sq F value Pr(>F)
## EDUC          10    14903  1490.3    50.98 <2e-16 ***
## Residuals    647443 18925824      29.2
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

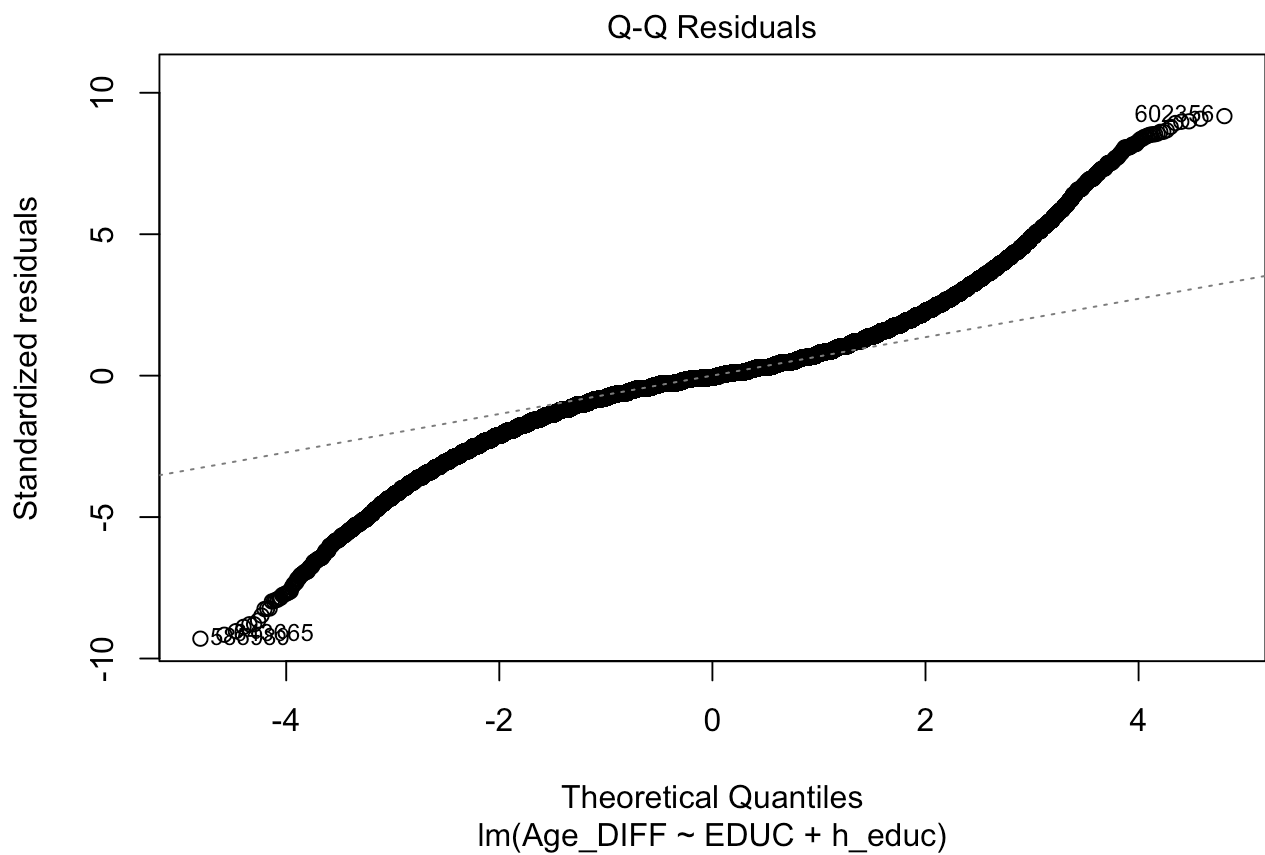
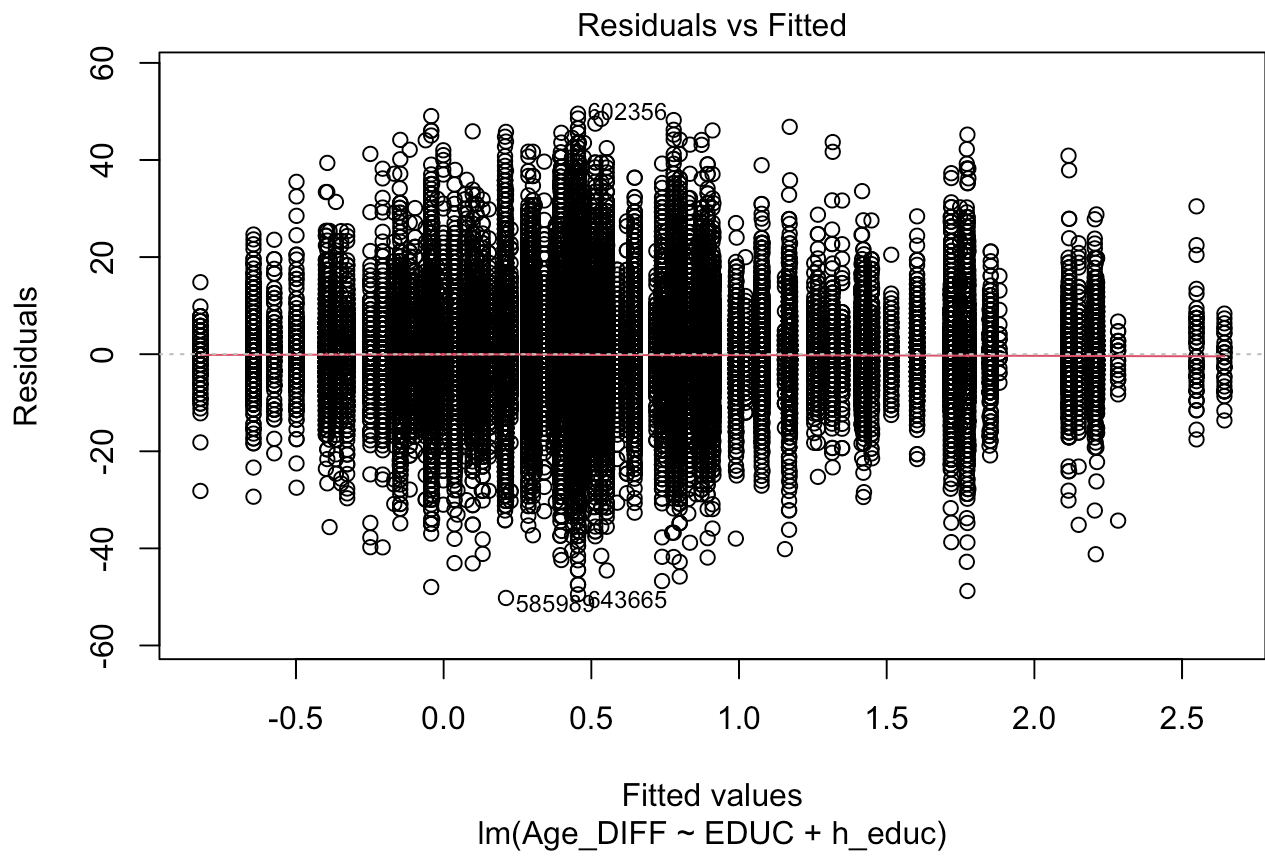
```
anova_H_EDUC = aov(Age_DIFF ~ h_educ, data = acs2021_couples_spouse)
summary(anova_H_EDUC)
```

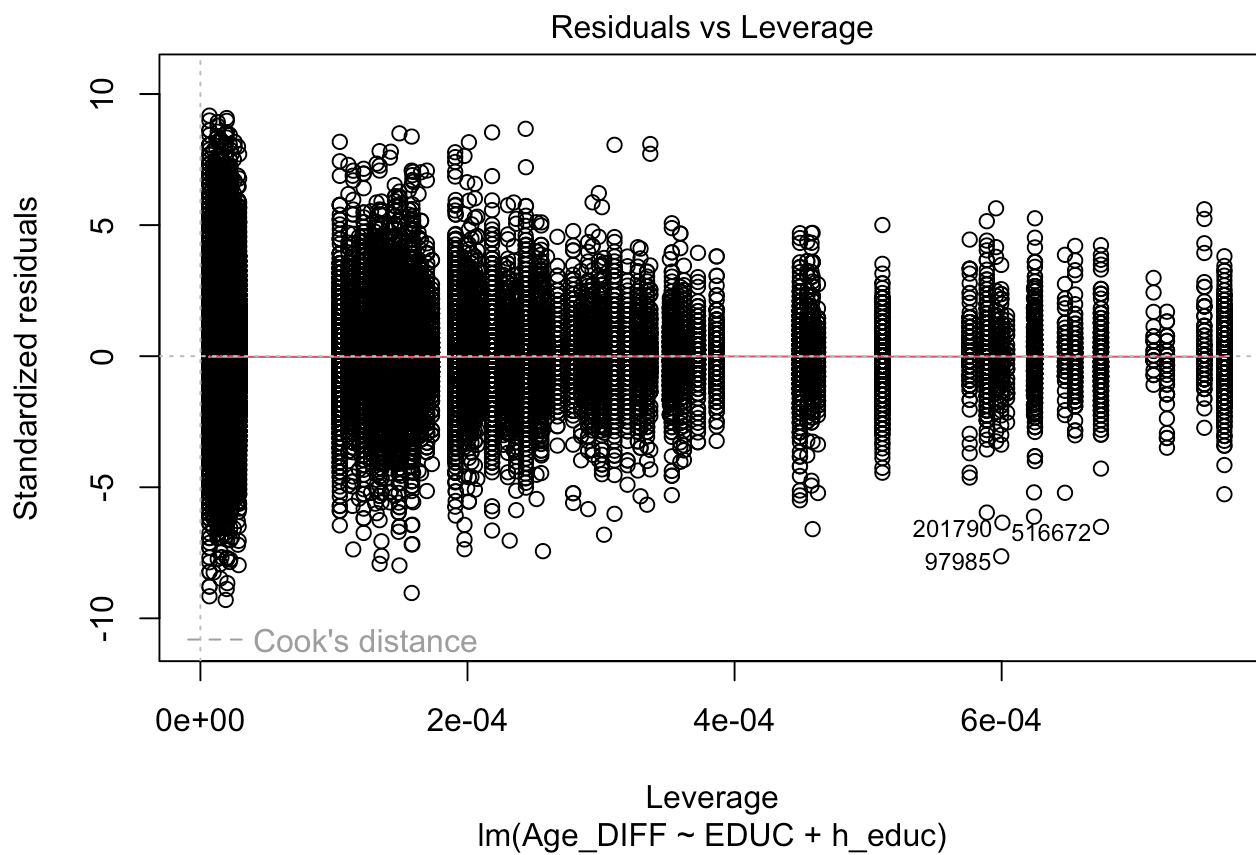
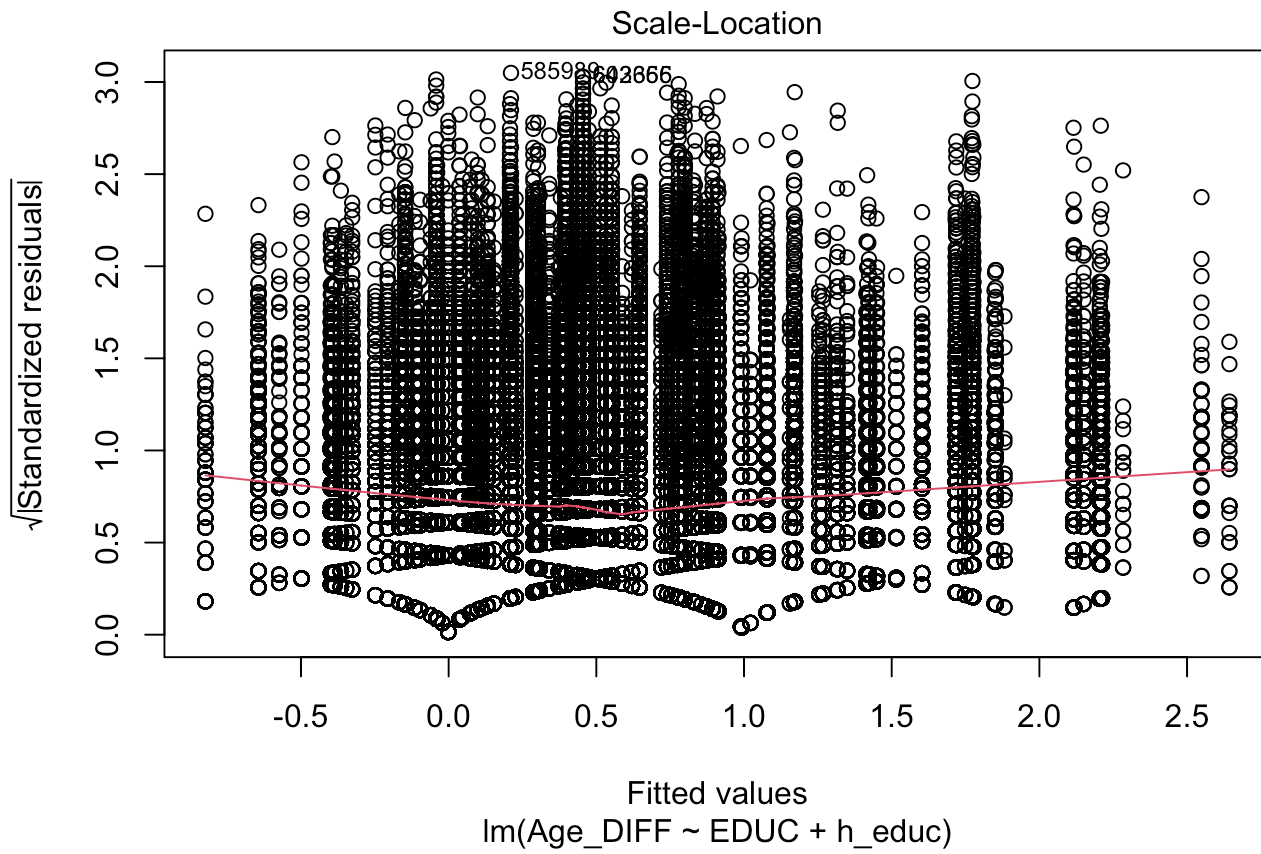
```
##              Df    Sum Sq Mean Sq F value Pr(>F)
## h_educ         10    37748   3775    129.3 <2e-16 ***
## Residuals    647443 18902979      29
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
lm <- lm(data=acs2021_couples_spouse, formula = Age_DIFF ~ EDUC + h_educ)
summary(lm)
```

```
##
## Call:
## lm(formula = Age_DIFF ~ EDUC + h_educ, data = acs2021_couples_spouse)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -50.211  -2.458  -0.302   2.487  49.545
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      1.718678    0.068027   25.265 < 2e-16 ***
## EDUCNursery school to grade 4  -0.726798    0.126449   -5.748 9.05e-09 ***
## EDUCGrade 5, 6, 7, or 8      -0.547083    0.078530   -6.967 3.25e-12 ***
## EDUCGrade 9                 -0.401408    0.093975   -4.271 1.94e-05 ***
## EDUCGrade 10                -0.301892    0.087687   -3.443 0.000576 ***
## EDUCGrade 11                -0.268148    0.085453   -3.138 0.001701 **
## EDUCGrade 12                 0.054614    0.059916    0.912 0.362022
## EDUC1 year of college         0.057823    0.062427    0.926 0.354320
## EDUC2 years of college        0.493412    0.063312    7.793 6.53e-15 ***
## EDUC4 years of college        0.398959    0.061386    6.499 8.08e-11 ***
## EDUC5+ years of college       0.133497    0.062465    2.137 0.032587 *
## h_educNursery school to grade 4  0.431267    0.144792    2.979 0.002897 **
## h_educGrade 5, 6, 7, or 8     -0.002089    0.089567   -0.023 0.981389
## h_educGrade 9                -0.696701    0.106580   -6.537 6.29e-11 ***
## h_educGrade 10               -0.862998    0.103519   -8.337 < 2e-16 ***
## h_educGrade 11               -0.944959    0.099656   -9.482 < 2e-16 ***
## h_educGrade 12               -1.318417    0.068225  -19.325 < 2e-16 ***
## h_educ1 year of college       -1.378064    0.070098  -19.659 < 2e-16 ***
## h_educ2 years of college      -1.815429    0.070898  -25.606 < 2e-16 ***
## h_educ3 years of college      -1.565444    0.069167  -22.633 < 2e-16 ***
## h_educ4 years of college      -1.338748    0.069817  -19.175 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.4 on 647433 degrees of freedom
## Multiple R-squared:  0.003319, Adjusted R-squared:  0.003288
## F-statistic: 107.8 on 20 and 647433 DF, p-value: < 2.2e-16
```

```
plot(lm)
```





```
lm <- lm(data=acs2021_couples_spouse, formula = Age_DIFF ~ EDUC + h_educ + born_in_USstate)
summary(lm)
```

```
##
## Call:
## lm(formula = Age_DIFF ~ EDUC + h_educ + born_in_USstate, data = acs2021_couples_spouse)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -50.049  -2.448  -0.236   2.533  49.202
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      1.973147    0.067896   29.061 < 2e-16
## EDUCNursery school to grade 4 -0.851896    0.126024  -6.760 1.38e-11
## EDUCGrade 5, 6, 7, or 8 -0.503437    0.078260  -6.433 1.25e-10
## EDUCGrade 9 -0.204854    0.093694  -2.186  0.0288
## EDUCGrade 10  0.099435    0.087587   1.135  0.2563
## EDUCGrade 11  0.180536    0.085417   2.114  0.0346
## EDUCGrade 12  0.517194    0.060103   8.605 < 2e-16
## EDUC1 year of college  0.550161    0.062640   8.783 < 2e-16
## EDUC2 years of college  0.968474    0.063487  15.255 < 2e-16
## EDUC4 years of college  0.844911    0.061532  13.731 < 2e-16
## EDUC5+ years of college  0.549516    0.062555   8.784 < 2e-16
## h_educNursery school to grade 4  0.277228    0.144308   1.921  0.0547
## h_educGrade 5, 6, 7, or 8  0.004627    0.089256   0.052  0.9587
## h_educGrade 9 -0.632922    0.106215  -5.959 2.54e-09
## h_educGrade 10 -0.606270    0.103230  -5.873 4.28e-09
## h_educGrade 11 -0.659622    0.099401  -6.636 3.23e-11
## h_educGrade 12 -1.005766    0.068147 -14.759 < 2e-16
## h_educ1 year of college -1.055993    0.070019 -15.082 < 2e-16
## h_educ2 years of college -1.493614    0.070814 -21.092 < 2e-16
## h_educ3 years of college -1.275980    0.069061 -18.476 < 2e-16
## h_educ4 years of college -1.096392    0.069669 -15.737 < 2e-16
## born_in_USstateborn in a state in the US -1.198556    0.017833 -67.211 < 2e-16
##
## (Intercept) ***
## EDUCNursery school to grade 4 ***
## EDUCGrade 5, 6, 7, or 8 ***
## EDUCGrade 9 *
## EDUCGrade 10
## EDUCGrade 11 *
## EDUCGrade 12 ***
## EDUC1 year of college ***
## EDUC2 years of college ***
## EDUC4 years of college ***
## EDUC5+ years of college ***
## h_educNursery school to grade 4 .
## h_educGrade 5, 6, 7, or 8
## h_educGrade 9 ***
## h_educGrade 10 ***
## h_educGrade 11 ***
## h_educGrade 12 ***
## h_educ1 year of college ***
```



```
## h_educ2 years of college          ***
## h_educ3 years of college          ***
## h_educ4 years of college          ***
## born_in_USstateborn in a state in the US ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.381 on 647432 degrees of freedom
## Multiple R-squared:  0.01022,    Adjusted R-squared:  0.01019
## F-statistic: 318.5 on 21 and 647432 DF,  p-value: < 2.2e-16
```

```
plot(lm)
```

